

Aryan Desai

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++

AI/ML: PyTorch, LangChain/LangGraph, RAG Pipelines, pgvector, HuggingFace, Reinforcement Learning, GCP Vertex AI, OpenAI API, Anthropic API

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend & Databases: Node.js, Express.js, PostgreSQL, Supabase, Drizzle ORM, Prisma, JWT

Tools & Platforms: Git, Docker, Temporal, GitHub Actions (CI/CD), Jest, Jira, Agile/Scrum, AWS, Web3

EXPERIENCE

AI Project Lead | Computing Councils of Canada (CCubed) | [GitHub](#)

March 2025 – Present

LangGraph, Docker, Python

Remote

- Scaled CCubed's pilot AI program to 10+ Canadian universities, growing reach to 500+ computing students through structured club partnerships
- Built an adversarial cybersecurity sandbox using LangGraph, orchestrating autonomous Red/Blue Team AI agents across 40 real-world CVEs, achieving ~85% average threat detection rate and reducing manual evaluation time by 60%
- Delivered an autonomous systems security research initiative end-to-end as tech lead, coordinating milestone delivery with the Director of AI on schedule

Frontend Developer & Workshop Lead | Google Developer Groups on Campus | [GitHub](#)

November 2025 – Present

React.js, TypeScript, GCP

Toronto, ON

- Designed and delivered 3+ technical workshops in React, TypeScript, and GCP, onboarding 30+ first-year students to full-stack and AI development
- Led frontend rebuild of the SolutionHacks hackathon website, shipping registration and resource-sharing features used by 200+ participants

Full Stack Developer Intern | FschoolAI | [GitHub](#)

May 2026 – Present

React.js, TypeScript, Python, Node.js

Remote

- Built a Chrome extension streamlining file upload workflows for students, reducing upload time by ~70% and eliminating 4+ manual steps per submission
- Contributed to design and development of 3+ AI agents including research, writing, and monitoring agents, improving platform automation coverage across core student workflows

Software Engineer Intern | FlipPilot | [GitHub](#)

February 2026 – May 2026

TypeScript, Temporal, PostgreSQL, Drizzle ORM

Remote

- Built a backend retail extraction pipeline with RESTful API integrations across 7+ US marketplaces via a 3-phase fetch strategy, cutting lookup time from ~5 min to under 30s (~90% reduction)
- Engineered an 8-stage automated workflow surfacing 50+ deals/day via AI-powered price analysis and multi-channel alerts, with all results persisted to a PostgreSQL database

PROJECTS

BeaverBuddy | [GitHub](#) | Next.js, TypeScript, PostgreSQL, Prisma, GPT-4o-mini

- **Won \$4,500 (1st/30+ teams)** at Build Your Bridge; built a full-stack mental health and immigration platform serving daily AI-generated quests to Canadian newcomers
- Integrated GPT-4o-mini with custom prompt engineering to power dynamic quest generation across 50+ immigration pathways, with JWT-authenticated role-based access

HomeCrowd | [GitHub](#) | Python, FastAPI, NVIDIA cuOpt, cuML, cuDF, Nemotron, MapLibre

- **Finalist at NVIDIA Hackathon**; built a three-agent AI pipeline (Pulse/Signal/Cue) ingesting live Toronto Open Data to generate 48-hour operational playbooks for small businesses, running fully local on NVIDIA GB10 Grace Blackwell with no cloud dependency
- Implemented GPU-accelerated staffing and inventory optimization using NVIDIA cuOpt's native `linear_programming` API with cuML `RandomForestRegressor` for per-business revenue forecasting, with automatic CPU fallback via PuLP/CBC

Disputr | [GitHub](#) | TypeScript, GOAT Network, x402, ERC-8004, LLM

- **Won 4 Awards (Market Master, x402 Master, UX Master, Security Master)** at OpenClaw AI Toronto Hackathon out of 500+ builders; built an autonomous chargeback defense agent for online sellers
- Engineered x402 payment integration on GOAT Network mainnet for autonomous dispute processing, with zero data retention and human-in-the-loop guardrails for high-risk actions

EchoBase | [GitHub](#) | Next.js, TypeScript, Express.js, PostgreSQL, Supabase, pgvector

- Built a workplace intelligence platform using a pgvector-backed semantic search layer and LLM reasoning to answer natural language queries ~3x faster than keyword search
- Designed a full-stack REST API backed by Supabase and PostgreSQL supporting knowledge base management, vector embeddings, and AI assistant functionality

EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Science, Computer Science Co-op

Sept. 2024 – June 2029

- Awarded \$6,000 Entrance Scholarship for strong academic performance (90–94 average)
- Dean's List, Faculty of Science – 2 consecutive years (2024/2025, 2025/2026), maintaining a 3.5+ GPA